

MELISSA LIZETH CONTRERAS ROJAS

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PROFILE

Software Engineer with experience in frontend and full-stack development, focused on scalable systems, performance, and reliable software delivery. Experience in high-impact environments such as Mercado Libre, working on system migrations, SQL optimization, and production support.

Background in UX and XR, including the development of a virtual reality solution using Meta Quest Pro. Published author in Springer on the use of virtual reality for cognitive assessment.

TECHNICAL SKILLS

Frontend development: React, Next.js, JavaScript, TypeScript, HTML, CSS, Flutter (Dart)

Backend development: Node.js, Java, Python, Golang, C#.

Databases: PostgreSQL, Firebase, MongoDB.

Tools: Git, Github, Docker, Postman, Bitbucket, Jira, Swagger.

Others: REST APIs, Architecture Principles, Debugging, Testing, Agile development, CI/CD environments, and collaborative software delivery.

PROJECTS

Developed software libraries for image processing, data analysis, and text processing using machine learning tools in collaborative environments. Participated in programming competitions focused on building efficient and scalable solutions.

PUBLICATION

Lead author of "Using Virtual Reality to Detect Memory Loss: An Exploratory Study" published in Springer Nature (Advances in Computing) and presented at the 17th Colombian Computing Conference (17CCC). Research focused on using virtual reality environments and interactive avatars to support early detection of cognitive impairment through behavioral analysis.

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LANGUAGES

English: B2 **Spanish** Native

EDUCATION

B.Sc. in Systems and Computing Engineering

Universidad de los Andes | 2020 - 2024

Relevant coursework: Software Engineering in Teams, Web Development, Mixed Reality Applications, Images and Vision, Product Design and Innovation, Data Structures and Algorithms, Algorithm Design and Analysis

EXPERIENCE

TUL | Software Engineer | Aug 2025 - Dec 2025

Developed features for web and mobile applications using Flutter (Dart) and Golang, improving performance and supporting scalable system growth. Contributed to code migrations and collaborated with product and design teams to deliver business-aligned solutions.

UBITS | Software Engineer | Feb 2025 - Jun 2025

Worked on the VALU platform, implementing new features and resolving production issues. Contributed to system stability and adaptation after acquisition, supporting continuous product evolution.

Mercado Libre | Software Intern Aug 2024 - Feb 2025

Optimized SQL queries and supported internal systems, ensuring reliability and efficient issue resolution. Participated in database and messaging migrations in production environments. Improved system understanding and operational efficiency through clear technical documentation.

Globalink Research Internship | Research Intern May 2024 - Jul 2024

Developed a virtual reality solution using Meta Quest Pro, integrating gesture tracking and real-time emotional feedback through interactive avatars.

Universidad de los Andes | Teaching Assistant for Data Structures and Algorithms Course Jan 2023 - Jun 2023

Supported Data Structures course, helping students solve complex problems and strengthen programming fundamentals.

Universidad de los Andes | Teaching Assistant for Introduction to Programming Course Aug 2020 - Nov 2020

Supported Introduction to Programming course, helping students solve complex problems and strengthen programming fundamentals.